

Do I Have Extra Sensory Perception?

Introduction

Extra Sensory Perception (ESP) is the ability to accurately predict what will happen using supernatural methods. Do I have it? If so, I could make lots of money by betting on the stock market! To find out, I took an online ESP test and then analyzed the data using the Binomial Test.

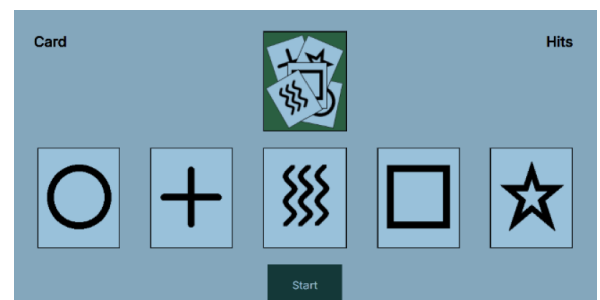
Let p be the probability that I have an accurate prediction. In the online ESP test, there were 5 possible predictions so if I don't have ESP, then $p = 1/5$. If I do have ESP, then $p > 1/5$. These hypotheses in conventional notation are:

$$H_0: p = 0.2 \text{ vs. } H_a: p > 0.2$$

Data Collection

I recorded the number of correct predictions out of 25 attempts in the online ESP test at [Advanced ESP card guessing \(Zener Cards\) test - PsychicScience.org](https://psychicscience.org/advanced-esp-card-guessing-zener-cards-test). I believe this data collection satisfied the 4 conditions of a Binomial Random Process:

1. Each prediction was correct or not (binary outcome).
2. I made a fixed number of attempts, $n = 25$ (fixed number of trials).
3. My probability of getting a prediction correct shouldn't change between attempts (fixed probability of success in a single trial).
4. I didn't use information from my previous attempts to inform my predictions (independent trials).



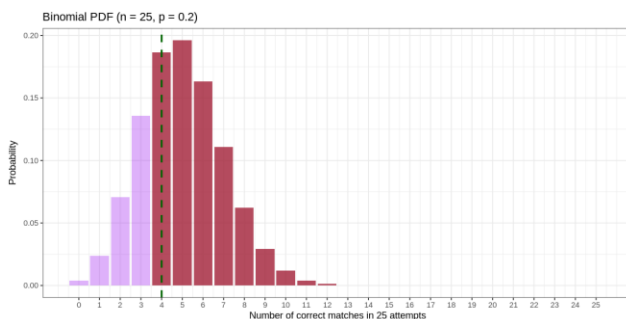
Data Analysis

Descriptive Statistics

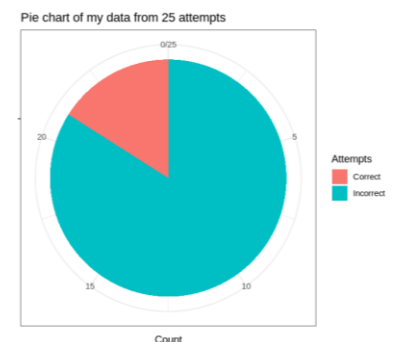
In 25 attempts, I correctly predicted the shape on 4 cards (16%).

Inferential Statistics

Let X be the number of correct predictions in 25 attempts. Assuming I'm just guessing, X is a Binomial Random Variable with $n = 25$ and $p = 0.2$.



I computed the probability of each value of X (PDF) using the `dbinom` function in R with `x=0:25`, `size=25`, `prob=0.2` (shown as purple bars) and then computed the probability of seeing my data (4) or more correct predictions using `pbinom` with `q=3` and `lower.tail=FALSE`. This p-value is 0.76 and corresponds to the brown area to the right of the vertical line at my data (green) in the plot of the binomial probabilities.



Conclusion

There is a 76% chance of seeing my results of 4 correct predictions or more in another set of 25 attempts if I don't have ESP and am just guessing. Since this is a large probability, I've concluded I don't have ESP and will find other ways besides betting to make money.